

Physics-Informed Neural Network Based Online Impedance Identification of Voltage Source Converters

Mengfan Zhang , Member, IEEE, Qianwen Xu , Member, IEEE, and Xiongfei Wang , Senior Member, IEEE

Abstract—The wide integration of voltage source converters (VSCs) in power grids as the interface of renewables causes the converter-grid interaction stability challenge. The black-box impedance of VSCs identified at the converter terminal is the key to facilitate the study of converter-grid interaction stability. However, since the limited impedance data amount in online measurement, the existing impedance identification methods cannot accurately capture characteristics of the impedance model in various operating scenarios with the changing profiles of renewables and loads. In this article, a physics-informed neural network based impedance identification is proposed to fill this research gap. The physics knowledge of the VSC is used to compress the artificial neural network, which can reduce the calculation burden of online impedance identification. Meanwhile, the two-steps impedance identification is developed with the inspiration of the transfer learning theory to further increase the online impedance identification efficiency. This method can significantly reduce the required data amount used in online impedance identification for the online stability analysis with the changing operating points. The case studies confirm the effectiveness of the proposed method.

Index Terms—Grid-converter interaction, online impedance identification, physics-informed neural network, renewables, transfer learning.

I. INTRODUCTION

VOLTAGE source converters (VSCs) are increasingly used as the interface of renewables into the grid, which makes the modern power grid more sustainable and efficient [1]. However, the interaction of the VSCs and grid imposes stability issues on the system [2], [3].

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Mengfan Zhang and Qianwen Xu are with Electrical Power and Energy Systems Division, KTH Royal Institute of Technology, 114 28 Stockholm, Sweden (e-mail: mezhang@kth.se; qianwenx@kth.se).

Xiongfei Wang is with the Department of Energy Technology, Aalborg University, 9220 Aalborg, Denmark (e-mail: xwa@et.aau.dk).

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The impedance-based stability analysis method is widely used for VSC-grid interaction systems since the method can predict the system stability based on the equivalent converter impedance and the grid impedance without modeling the whole system [4], [5]. A typical application is reported in [6], an analytical impedance model of the grid-connected VSC is derived, and then the generalized Nyquist criterion (GNC) is used to analyze the stability of grid-converter interaction system with the equivalent grid impedance and the derived VSC impedance model.

However, since limited information on the VSC control system is provided by the vendors, it is hard to derive the analytical impedance model of VSC in practical application. The impedance measurement based stability analysis is thus increasingly applied in practical application without intellectual property contamination. An impedance measurement based stability analysis is reported in [7], the measured VSC impedance and the known grid impedance are used for the stability analysis of the grid-converter system. Yet, the existing impedance measurements are conducted only at the fixed operating point. Since the impedance of converters changes along with the operating point, the obtained impedance at few operating points cannot be used for stability analysis when the operating point of the VSC is varying continually [8]. To analyze the stability of the grid-converter interaction system considering continuous fluctuations of renewables and loads in the system, there is always a demand for the online impedance identification of VSC with the changing operating points.

To address this issue, a few methods are reported in previous literature. A polytopic modeling method for power-electronics converters is reported in [9], where the small-signal models at different operating points are summarized with respective weights to generate the multioperating-point (MOP) model. Yet this model can just predict the model of the converters at very limited operating points and the accuracy is hard to be guaranteed. A grey box method is developed in [10] by identifying control parameters of VSC, but the limitation is that the control structure needs to be known. An identification method is proposed in [11] to derive the impedance model with wide operating conditions, but it is still limited to a known structure or its equivalent structure (from dq to abc) while it cannot be extended to VSC with a different control structure or with additional control blocks. To enable the impedance identification of unknown

structure, an improved method is to adapt the artificial neural network (ANN) technique to identify the impedance model of the VSC at different operating points [12]. The ANN-based MOP-impedance model is trained with the measured impedance, which can predict the accurate impedance of the VSC at varying operating points. However, the accuracy of the trained model depends on the data amount used in model training, in online impedance identification, the limited measured data amount will consequently introduce the overfitting phenomena to the model, and then affect the accuracy of stability analysis [13]. A transfer learning based method in [14] is proposed to accelerate the training process using simulation data, which requires further analytical analysis and experimental results for verification. Moreover, for all aforementioned methods, impedance models are obtained by offline simulation/experiments while the online impedance identification is not studied, which is significantly important for the online stability evaluation under the changing operating profiles.

To overcome the bottleneck in online impedance identification, this article proposes a physics-informed neural network based impedance identification method. The basic idea is to use the physical knowledge of VSC to facilitate the black-box modeling. Following this concept, we propose a new physics-informed neural network based online impedance identification method that if a neural network can be constructed using the physical model of the general VSC, the accuracy of the trained models can be improved even with limited data. Meanwhile, to improve the efficiency of online impedance identification, the transfer learning technique is integrated into the method by transferring offline knowledge to online impedance identification [15]. The proposed method consists of the following steps: First, the ANN model is constructed based on the derived analytical impedance model of the general VSC. Next, offline tests are conducted to generate enough impedance data, which are fed into the constructed ANN to generate the offline impedance model. Then, based on transfer learning theory, the offline impedance model is transferred to online impedance identification of different VSCs by minimizing the distribution differences between the trained data and limited online field measured data iteratively. At last, the online impedance model is generated. The main contributions include the following.

- 1) The physics of VSC is used to compress the ANN structure to reduce the demand for data and training time, to achieve fast and accurate online identification.
- 2) The online impedance identification of a target VSC with the unknown control structure is achieved by using a general VSC simulation model and the online measured data.

The rest of this article is organized as follows. The framework of the proposed physics-informed neural network based online impedance identification of VSC is stated in Section II. In Section III, the implementation of the proposed method is further explained. Comparative case studies are provided in Section IV that validate the effectiveness of the proposed method. Finally, Section V concludes this article.

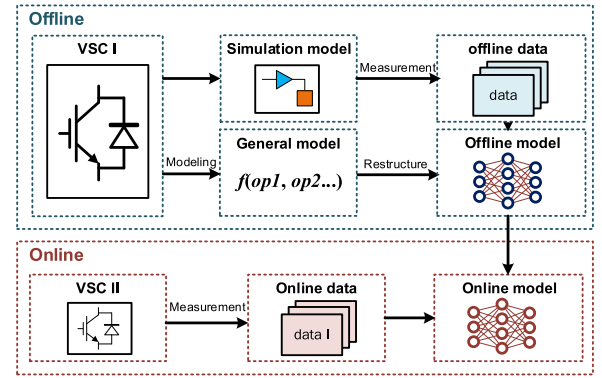


Fig. 1. Framework of the proposed physics-informed neural network based online impedance identification method.

II. FRAMEWORK OF THE PROPOSED METHOD

The black-box impedance of VSCs identified at the converter terminal is the key to facilitate the study of converter-grid interaction stability. In real industrial practice, vendors are unwilling to share the inner control structure and parameters. But the control objective of VSCs is known, and it is based on the general feedback control structure with standard circuits. Therefore, the idea of the proposed method is to utilize partial physics of the VSC and a small amount of online measured data to identify the accurate impedance model of VSC. The preliminary impedance model is generated at the offline phase to reduce the online training time and the data amount requirement.

The framework of the proposed physics-informed neural network based online impedance identification is shown in Fig. 1. Here, it should be mentioned that the target VSC to be identified has an unknown control structure and unknown filter structure, and we will use the general VSC to generate data for offline training and use online measured data for the online identification of the target VSC. The implementation of the method is divided into the offline phase and the online phase.

In the offline phase, first, the theoretical model of general VSC that is derived from the physics of the VSC is used to restructure the ANN structure, which can compress the ANN structure and, thus, reduce the demand for data amount and training time. Second, the offline impedance data are gained through the simulation of the general VSC. Then, the offline MOP-impedance model is generated with the training by feeding the offline impedance data into the structured ANN model.

In the online phase, with the transfer learning technique, the field measured impedance data of the VSC are fed into the generated offline MOP-impedance model. After the online training, the online MOP-impedance model of the VSC is achieved.

III. IMPLEMENTATION OF PHYSICS-INFORMED NEURAL NETWORK BASED ONLINE IMPEDANCE IDENTIFICATION

A. Offline Phase

In the offline phase, first, the theoretical model of general grid-connected VSC system is derived, then an ANN is structured based on the derived model. After that, the structured

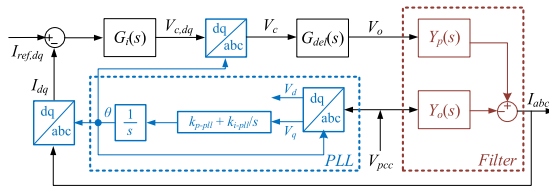
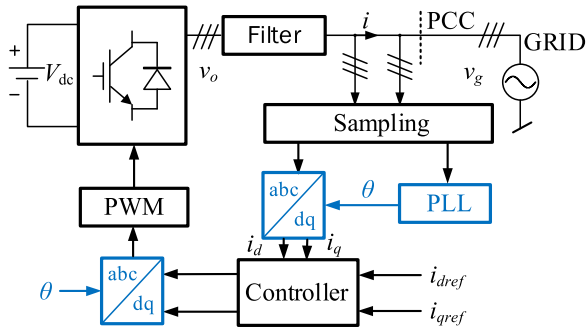
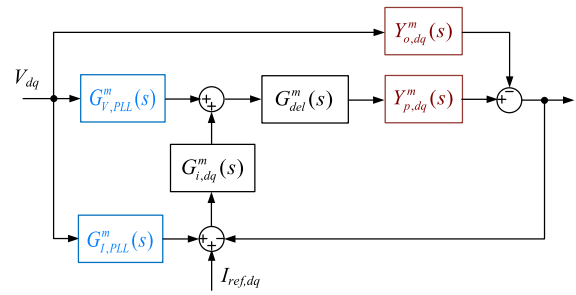


Fig. 3 shows the detailed control diagram of the general grid-connected VSC. $G_i(s)$ is the transfer function of the current controller, $Y_p(s)$ and $Y_o(s)$ are the transfer function of the filter. $G_{del}(s)$ is the time delay introduced by the digital control system [16], which consists of the computational delay (T_s) and pulsewidth modulator delay ($0.5T_s$), where T_s is the sampling period. The time delay is shown as follows:

The synchronous reference frame PLL is linearized as follows [17]:

$$G_{\text{PLL}}(s) = \frac{s \cdot k_{p-\text{pll}} + k_{i-\text{pll}}}{s^2 + (s \cdot k_{p-\text{pll}} + k_{i-\text{pll}}) V_d} \quad (3)$$

$$= (I + \Delta I) \cos(\Delta\theta + \theta) + i \cdot (I + \Delta I) \sin(\Delta\theta + \theta)$$

$$\rightarrow \Delta V_{\text{PLL},dq} = i \cdot \Delta\theta \cdot V_{s,dq} \quad (5)$$
$$\Delta V_{\text{PLL},dq} = i \cdot G_{\text{PLL}}(s) \Delta V_q \cdot V_{s,dq}. \quad (6)$$
$$G_{\text{del}}^m(s) = \begin{bmatrix} G_{\text{del}}(s) & 0 \\ 0 & G_{\text{del}}(s) \end{bmatrix} \quad (8)$$
$$\begin{aligned} \begin{bmatrix} \Delta I_{\text{PLL},d} \\ \Delta I_{\text{PLL},q} \end{bmatrix} &= \begin{bmatrix} 0 & -G_{\text{PLL}}(s) I_{s,q} \\ 0 & G_{\text{PLL}}(s) I_{s,d} \end{bmatrix} \begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix} = G_{I,\text{PLL}}^m \begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix} \\ \begin{bmatrix} \Delta V_{\text{PLL},d} \\ \Delta V_{\text{PLL},q} \end{bmatrix} &= \begin{bmatrix} 0 & -G_{\text{PLL}}(s) V_{s,q} \\ 0 & G_{\text{PLL}}(s) V_{s,d} \end{bmatrix} \begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix} = G_{V,\text{PLL}}^m \begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix} \end{aligned} \quad (9)$$
$$\begin{aligned} \begin{bmatrix} I_d \\ I_q \end{bmatrix} &= G_{cl,dq}^m(s) \begin{bmatrix} I_{\text{ref},d} \\ I_{\text{ref},q} \end{bmatrix} \\ &- \left\{ [I^m + T_{dq}^m(s)]^{-1} Y_{to,dq}^m(s) - G_{cl,dq}^m(s) Y_{\text{PLL}}^m(s) \right\} \\ &\times \begin{bmatrix} V_d \\ V_q \end{bmatrix} \end{aligned} \quad (10)$$

where I_m is the unitary diagonal matrix, the other transfer functions are shown as follows:

$$\begin{aligned} Y_{to,dq}^m(s) &= Y_{o,dq}^m(s) - Y_{p,dq}^m(s) G_{del}^m(s) G_{V,PLL}^m(s) \\ T_{dq}^m(s) &= Y_{p,dq}^m(s) G_{del}^m(s) G_{i,dq}^m(s) \\ G_{cl,dq}^m(s) &= [I^m + T_{dq}^m(s)]^{-1} T_{dq}^m(s). \end{aligned} \quad (11)$$

Thus, the closed-loop admittance matrix of VSC is derived as

$$Y_{tcl,dq}^m(s) = [I^m + T_{dq}^m(s)]^{-1} Y_{to,dq}^m(s) - G_{cl,dq}^m(s) Y_{PLL}^m(s). \quad (12)$$

As shown in (12), the converter admittance model is dependent on the steady-state operating point, frequency, and control parameters.

2) ANN Restructuration: The ANN training requires a large amount of data to configure the ANN structure and the weights of the connection of each neuron. If some parts of the ANN can be predecided, the demand for data and training time can be greatly reduced. Based on the derived theoretical admittance model of VSC, this section shows how to adopt this theoretical model to compress neural networks for online impedance identification.

The admittance model in (12) can be transformed as follows:

$$Y_{dq}^m(s) = [O_{dq}^m(s) - P_{dq}^m(s) \cdot V_{dq}^m - Q_{dq}^m(s) \cdot I_{dq}^m] \cdot [R_{dq}^m(s)]^{-1} \quad (13)$$

where

$$\begin{aligned} O_{dq}^m(s) &= Y_{o,dq}^m(s) \\ P_{dq}^m(s) &= \begin{bmatrix} 0 & -G_{PLL}(s) \\ 0 & G_{PLL}(s) \end{bmatrix} \cdot G_i(s) \cdot Y_{p,dq}^m(s) \\ Q_{dq}^m(s) &= \begin{bmatrix} 0 & -G_{PLL}(s) \\ 0 & G_{PLL}(s) \end{bmatrix} G_{del}(s) \cdot G_i(s) \cdot Y_{p,dq}^m(s) \\ R_{dq}^m(s) &= \begin{bmatrix} 1 + Y_p^{dd}(s) \cdot G_{del}(s) \cdot G_i(s) & \\ Y_p^{dq}(s) \cdot G_{del}(s) \cdot G_i(s) & \\ Y_p^{qd}(s) \cdot G_{del}(s) \cdot G_i(s) & 1 \\ +Y_p^{qq}(s) \cdot G_{del}(s) \cdot G_i(s) & \end{bmatrix} \\ V_{s,dq}^m &= \begin{bmatrix} 0 & V_{s,d} \\ 0 & V_{s,q} \end{bmatrix}, I_{s,dq}^m = \begin{bmatrix} 0 & I_{s,d} \\ 0 & I_{s,q} \end{bmatrix}. \end{aligned} \quad (14)$$

As shown in (13) and (14), the admittance model is naturally separated into the operating point relevant function and the frequency relevant function. For simplification, (14) can be rewritten as

$$\begin{aligned} O_{dq}^m(s) &= \begin{bmatrix} O_{dd}(s) & O_{dq}(s) \\ O_{qd}(s) & O_{qq}(s) \end{bmatrix} \\ P_{dq}^m(s) &= \begin{bmatrix} P_{dd}(s) & P_{dq}(s) \\ P_{qd}(s) & P_{qq}(s) \end{bmatrix} \\ Q_{dq}^m(s) &= \begin{bmatrix} Q_{dd}(s) & Q_{dq}(s) \\ Q_{qd}(s) & Q_{qq}(s) \end{bmatrix} \\ [R_{dq}^m(s)]^{-1} &= \begin{bmatrix} R_{dd}^{-1}(s) & R_{dq}^{-1}(s) \\ R_{qd}^{-1}(s) & R_{qq}^{-1}(s) \end{bmatrix}. \end{aligned} \quad (15)$$

Thus, (13) is transformed as

$$\begin{aligned} Y_{dd}(s) &= O_{dd}(s) R_{dd}^{-1}(s) + R_{qd}^{-1}(s) \cdot \\ &\quad (O_{qd}(s) - (V_q \cdot P_{dq}(s) + V_d \cdot P_{dd}(s)) \\ &\quad - (I_q \cdot Q_{dq}(s) + I_d \cdot Q_{dd}(s))) \\ Y_{qd}(s) &= O_{qd}(s) R_{dd}^{-1}(s) + R_{qd}^{-1}(s) \cdot \\ &\quad (O_{qq}(s) - (V_d \cdot P_{qd}(s) + V_q \cdot P_{qq}(s)) \\ &\quad - (I_d \cdot Q_{qd}(s) + I_q \cdot Q_{qq}(s))) \\ Y_{dq}(s) &= O_{dd}(s) R_{dq}^{-1}(s) + R_{qq}^{-1}(s) \cdot \\ &\quad (O_{qd}(s) - (V_q \cdot P_{dq}(s) + V_d \cdot P_{dd}(s)) \\ &\quad - (I_q \cdot Q_{dq}(s) + I_d \cdot Q_{dd}(s))) \\ Y_{qq}(s) &= O_{qd}(s) R_{dq}^{-1}(s) + R_{qq}^{-1}(s) \cdot \\ &\quad (O_{qq}(s) - (V_d \cdot P_{qd}(s) + V_q \cdot P_{qq}(s)) \\ &\quad - (I_d \cdot Q_{qd}(s) + I_q \cdot Q_{qq}(s))). \end{aligned} \quad (16)$$

Take Y_{dd} as example

$$\begin{aligned} Y_{dd}(s) &= F_1(s) - V_q \cdot F_2(s) - V_d \cdot F_3(s) - I_q \cdot F_4(s) - I_d \cdot F_5(s) \\ &= G[V_q, V_d, I_q, I_d, F_1(s), F_2(s), F_3(s), F_4(s), F_5(s)]. \end{aligned} \quad (17)$$

The magnitude and phase of the admittance model are written as follows:

$$\begin{aligned} Y_{dd-mag} &= \text{Mag}[G[V_q, V_d, I_q, I_d, F_1(s), \\ &\quad F_2(s), F_3(s), F_4(s), F_5(s)]] \\ Y_{dd-pha} &= \text{Ang}[G[V_q, V_d, I_q, I_d, F_1(s), \\ &\quad F_2(s), F_3(s), F_4(s), F_5(s)]] \end{aligned} \quad (18)$$

where Mag and Ang are the magnitude and phase calculation function, respectively.

In a conclusion, based on the analytical impedance model of the VSC, the ANN can be structured, as shown in Fig. 5. This structured neural network consists of two hidden layers and one calculation layer. Layer I is corresponding to the functions $F_1(s)$, $F_2(s)$, $F_3(s)$, $F_4(s)$, and $F_5(s)$, where the input is frequency ($s = j2\pi f$). Layer II is corresponding to the function G . Layer III is corresponding to the function Mag and Ang .

B. Online Phase

In the online phase, first the online impedance measurement is adopted to acquire the online impedance data. Then, the transfer learning technique is used to train the online MOP-impedance model by feeding the online measured data into the offline impedance model.

The online impedance measurement method is developed in variable operating points and frequency to obtain the dataset considering the variation of the operating points in different frequencies.

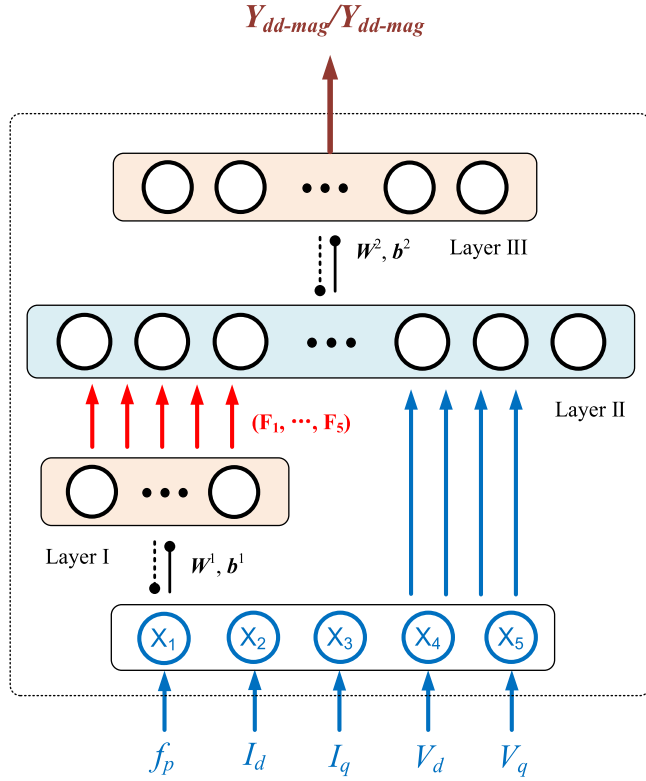


Fig. 5. Restructured ANN diagram.

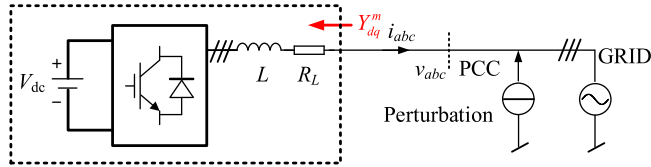


Fig. 6. Impedance measurement diagram.

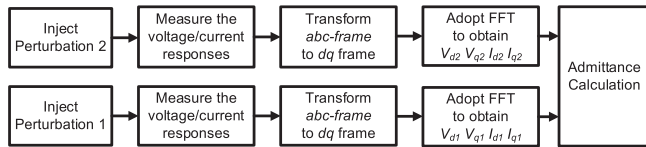


Fig. 7. Impedance measurement workflow.

Fig. 6 shows the diagram of dq -impedance measurement of the VSC. First, the current perturbation is injected into the VSC at the PCC. Second, the current and voltage response of the VSC is measured at the PCC. At last, the output impedance of VSC is calculated. The detailed workflow of the online impedance measurement is shown in Fig. 7.

First, the current perturbation at the specific frequency f_p is injected into the VSC system at PCC. Before the injection, the perturbation signal magnitude needs to be appropriately designed [18]. If the magnitude of the perturbation signal is not small enough, the operating point of the VSC will be changed. Meanwhile, if it is not sufficiently large, the noise disturbances

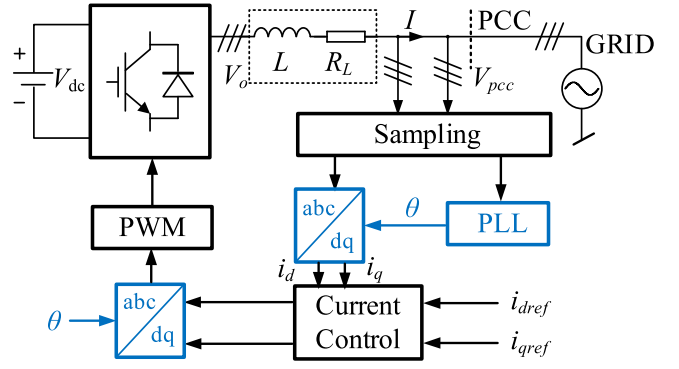


Fig. 8. Diagram of PI-controlled grid-connected VSC system.

will affect the measurement. According to the works in [19] and [20], the magnitude of the current perturbation is set to 5% of the operating point in this article.

Second, the measured voltage and current of the VSC at PCC are calculated to obtain the impedance. The Park's transformation are used to transform the abc -frame signal to dq -frame, i.e., V_{d1} , V_{q1} , I_{d1} , and I_{q1} . V_{d1} , V_{q1} , I_{d1} , and I_{q1} are the time series signal with the fixed sampling frequency. The frequency domain voltage and current signal at the injected frequency can be extracted with the fast Fourier transform algorithm, i.e., $V_{d1}(f_p)$, $V_{q1}(f_p)$, $I_{d1}(f_p)$, and $I_{q1}(f_p)$. To achieve the impedance matrix $Y_{dq}^m(f_p)$, two independent perturbation signals are injected to gather enough information to solve the impedance matrix [21]. With the voltage and current of the second injection, i.e., $V_{d2}(f_p)$, $V_{q2}(f_p)$, $I_{d2}(f_p)$, and $I_{q2}(f_p)$, the admittance of the VSC at specific frequency f_p is calculated as follows:

$$Y_{dq}^m(f_p) = \begin{bmatrix} Y_{dd}(f_p) & Y_{dq}(f_p) \\ Y_{qd}(f_p) & Y_{qq}(f_p) \end{bmatrix} \\ = \begin{bmatrix} V_{d1}(f_p) & V_{d2}(f_p) \\ V_{q1}(f_p) & V_{q2}(f_p) \end{bmatrix}^{-1} \begin{bmatrix} I_{d1}(f_p) & I_{d2}(f_p) \\ I_{q1}(f_p) & I_{q2}(f_p) \end{bmatrix}. \quad (19)$$

After the procedure, the admittance of the VSC at one operating point is generated. Repeating this procedure by changing the frequency of the perturbation and operating point of VSC, the admittance dataset for training is established. The steady-state operating points of the VSC, i.e., V_d , V_q , I_d , and I_q , are changing in different operating scenarios. Moreover, to reveal the characteristics of the VSC at the wideband frequency, the frequency f_p is selected as the input of the model. Thus, the structure of MOP-impedance model is established, where the operating point of the VSC in dq -frame, i.e., (V_d, V_q, I_d, I_q) , and frequency f_p are selected as the input of the model to reveal the operating-point-dependent and frequency-dependent feature. The magnitude and the phase of impedance in the dq -frame are selected to represent the outputs of the model.

Based on transfer learning theory, the domain data consist of a feature space of input X and a marginal probability distribution $P(X)$. Denoted by $D_s = \{(x_{s1}, y_{s1}), \dots, (x_{sn}, y_{sn})\}$ the source-domain labeled data, where the input X_s are operating points and frequency of the system, i.e., (f, V_d, V_q, I_d, I_q)

TABLE I
VSC PARAMETERS

Symbol	Description	VSC I Value	VSC II Value
V_{dc}	DC voltage of VSC	700 V	700 V
f_{sw}	Switching frequency	20 kHz	10 kHz
f_0	Fundamental frequency	50 Hz	50 Hz
L_f	Inverter inductor	1 mH	3 mH
R_{Lf}	Parasitic resistance of inverter inductor	0.3 mΩ	0.9 mΩ
K_{p_i}	Proportional parameter of the current controller	10.5	31.4
K_{i_i}	Integral parameter of the current controller	5900	16449
K_{p_pll}	Proportional parameter of the PLL	1.2	1.4
K_{i_pll}	Integral parameter of the PLL	257	324

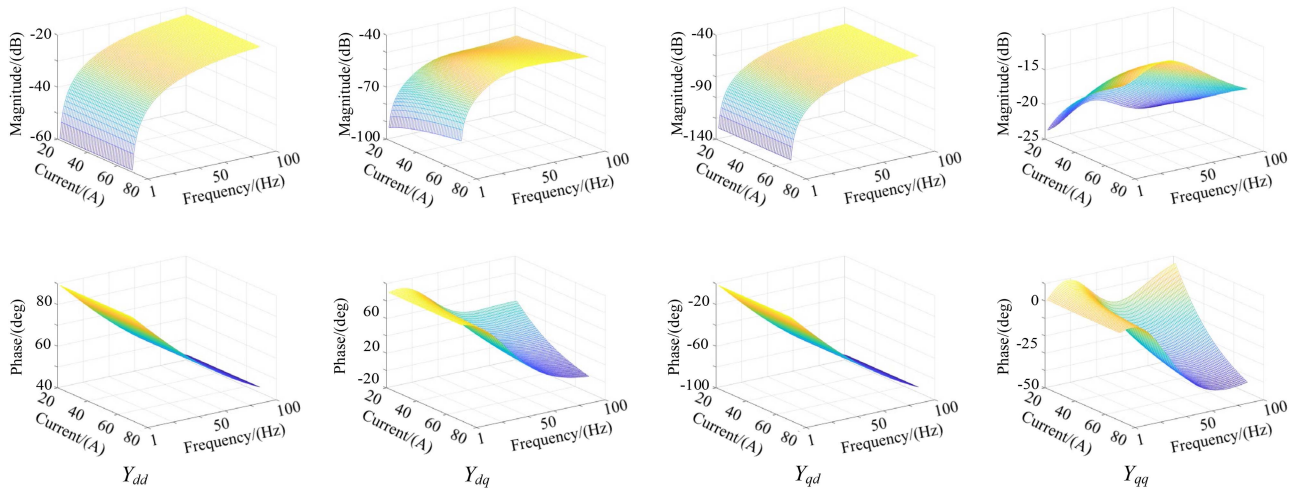


Fig. 9. Measured admittance dataset of VSC I.

and y_s are the corresponding admittance, i.e., $(Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq})$. Similarly, the field-measured impedance data in the target domain are represented as $D_t = \{(x_{t1}, y_{t1}), \dots, (x_{tm}, y_{tm})\}$, where $m < n$. In the transfer learning setting, if the data in the source and target domain follow the same distribution, i.e., $X_s \neq X_t, P(X_s) = P(X_t)$, the transferred admittance model can achieve a satisfactory application accuracy [22]. Obviously, the distribution of the online measured data and the offline training data can be easily guaranteed the same distribution in this article.

Thus, the online measured impedance data are fed into the pre-trained ANN, after several times training, the online impedance model is obtained.

IV. CASE STUDY AND VALIDATION

To validate the proposed physics-informed neural network based online impedance identification method, two case studies are designed. The first case study is designed to show the

advantages against the conventional ANN-based impedance identification method, and the comparison of different methods in impedance identification of PI-controlled grid-connected converter is demonstrated. The second case study is designed to show that the proposed method can be used to identify the impedance of grid-connected VSC with unknown control structures. Furthermore, the experiment is conducted to show the application of generated impedance model using the proposed method.

A. Applied in PI-Controlled Grid-Connected VSC and Comparison With the Conventional ANN Method

A target three-phase VSC system is marked as VSC II with the PI current controller and the PLL shown in Fig. 8. A known VSC is designed with the same control loop but different parameters and is marked as VSC I. The detailed parameters of the two

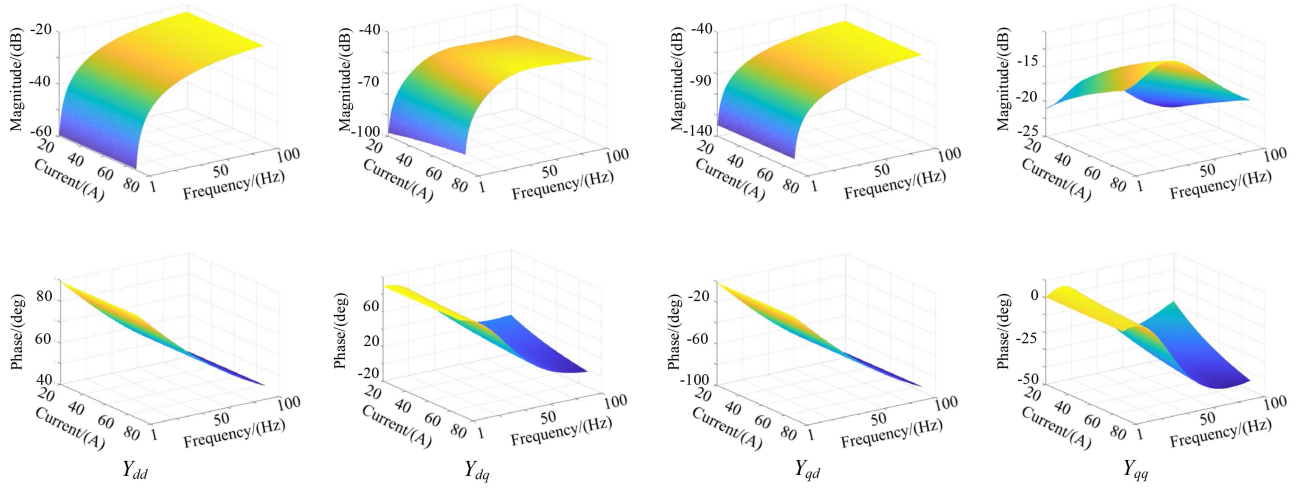


Fig. 10. Generated MOP-admittance model of VSC I.

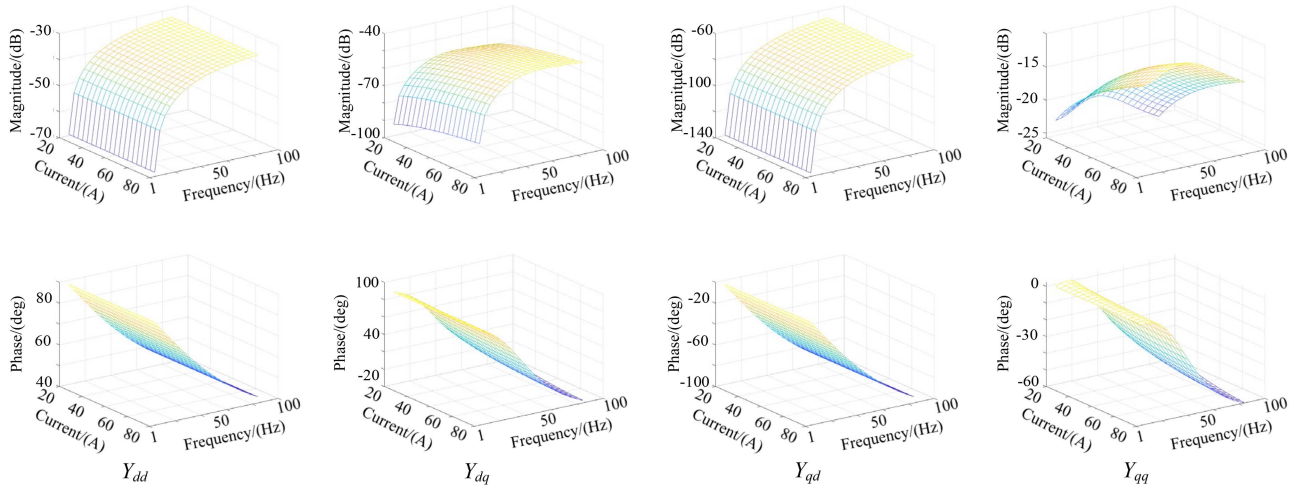


Fig. 11. Measured admittance dataset of VSC II.

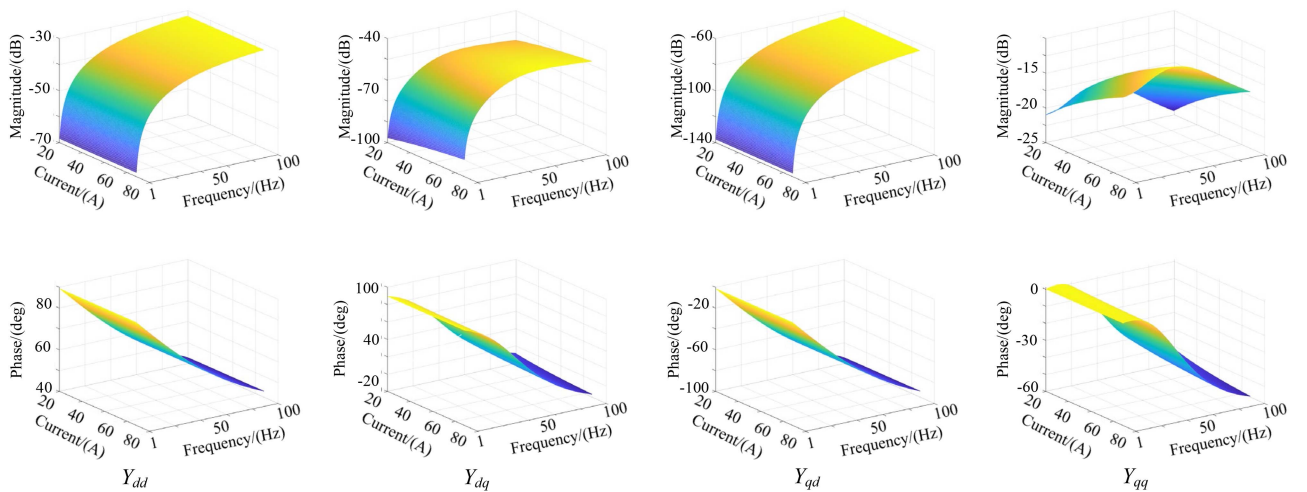


Fig. 12. Generated MOP-admittance model of VSC II.

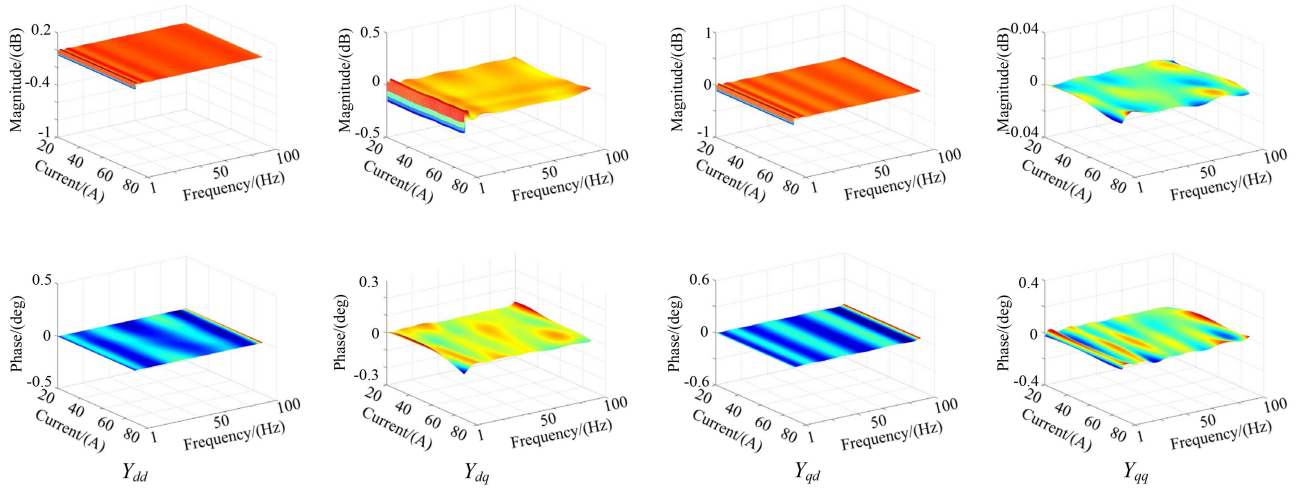


Fig. 13. Errors between generated admittance model using the proposed method and measured data.

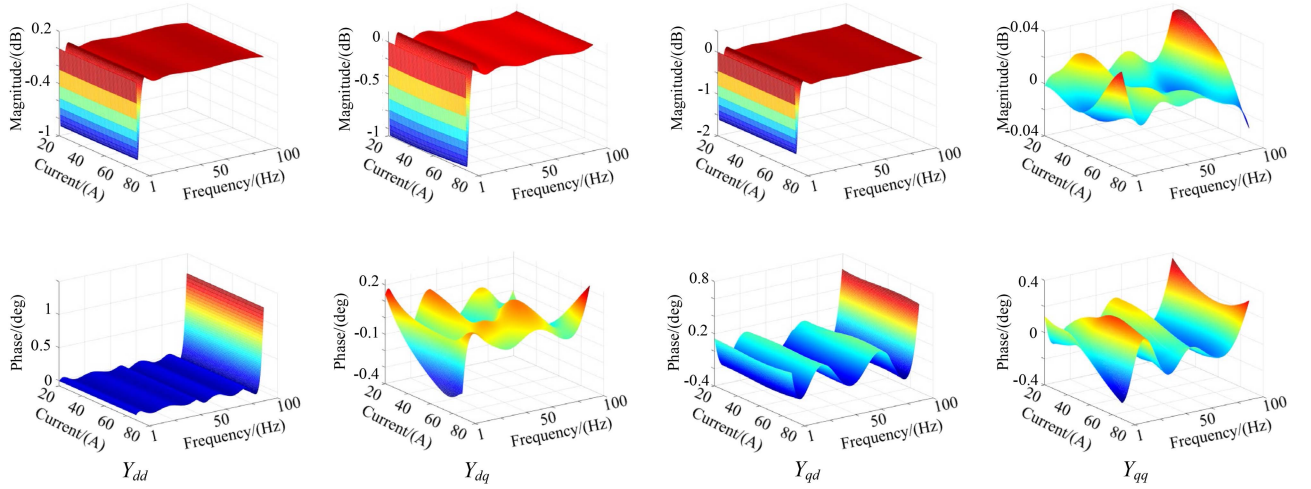


Fig. 14. Errors between generated admittance model using conventional ANN method and measured data.

VSCs are shown in Table I. In the case study setup, the VSC II is totally a black box while the VSC I is already known.

The task is to train the impedance model of VSC II using the known physics of VSC I and the limited online measured impedance of VSC II. Finally, to show the effectiveness of the proposed method, the comparison between the conventional ANN-based identification method and the proposed method using the same measured impedance dataset is conducted.

Before training the impedance model of the VSC II, the impedance model of VSC I need to be generated. In this article, the data used for offline impedance model training are extracted by the simulation model, which is validated by the field test. To make the result easily visible, only the variation of the d -axis current I_d is shown. The frequency f_p is from 1 to 100 Hz with 1-Hz interval, and the d -axis current I_d is from 20 to 70 A with 1-A interval. The measured admittance dataset is shown in Fig. 9. Then, the measured dataset is fed into the restructured ANN and the model is trained. At last, the admittance model of the VSC I Y_{VSC-I} is generated, as shown in Fig. 10.

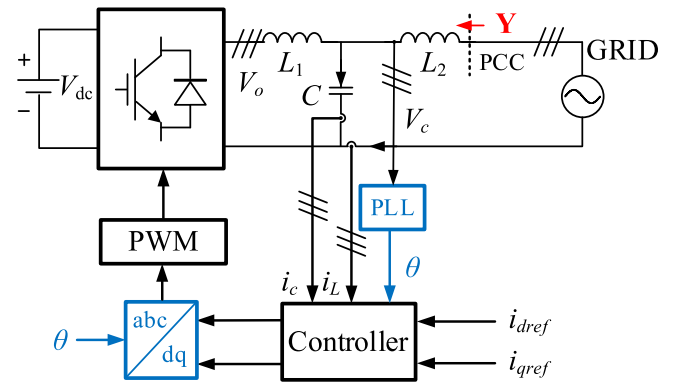


Fig. 15. Diagram of active damping control of LCL-filtered VSC grid-connected VSC system.

The model of VSC I is used to train the admittance model of the VSC II. In the impedance dataset of VSC II, the frequency

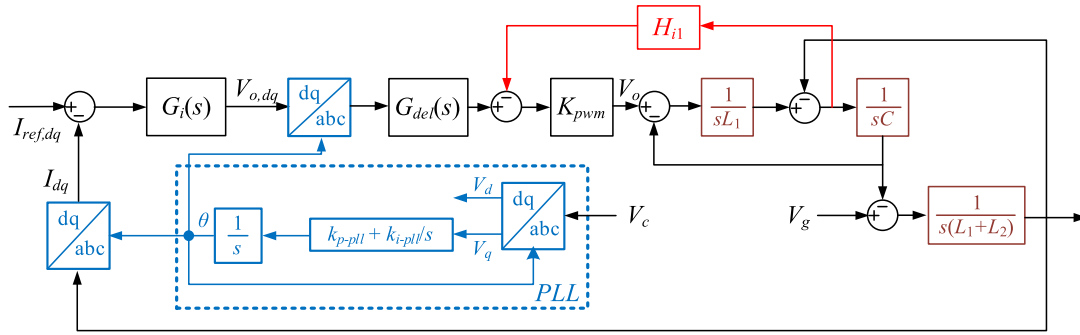


Fig. 16. Block diagram of active damping control of grid-connected converter.

f_p is from 1 to 101 Hz with 5-Hz interval, and the d -axis current I_d is from 20 to 70 A with 5-A interval. In the online phase, only 200 data points are measured, which is shown in Fig. 11. Then, the dataset is fed into the trained model Y_{VSC-I} with the transfer learning technique. The admittance model of the VSC II Y_{VSC-II} is generated, which is shown in Fig. 12.

To validate the effectiveness of the proposed method, the trained admittance model Y_{VSC-II} is compared with the field measured data. In the field measured dataset, the corresponding frequency f_{p-v} is from 1 to 100 Hz with 0.1-Hz interval, whereas the corresponding d -axis current I_{d-v} is from 20 to 70 A with 0.1-A interval. The corresponding errors are shown in Fig. 13, which are less than 1%. Hence, the comparison shows that the proposed method is effective in the online impedance identification of the VSC.

The comparison between the conventional ANN method and the proposed method using the same measured impedance dataset is conducted to show the effectiveness of the proposed method. The same dataset, as shown in Fig. 11, are fed into a two-hidden-layer random initialized ANN. After the training, the model $Y_{VSC-II-O}$ is generated. The corresponding errors between the verification data with the same distribution and the model generated by the conventional ANN method are shown in Fig. 14.

Compared to the errors shown in Fig. 13, the conventional ANN method cannot have a good performance with the sparse dataset, whereas the proposed method can accurately achieve the impedance model of VSC even with limited data amount.

B. Applied in Active Damping Controlled LCL-Type Grid-Connected VSC

To show the effectiveness of the proposed method to identify VSCs with unknown inner control structures, the proposed method is implemented to identify the impedance of a grid-connected VSC with LCL filter and active damping control, which is shown in Fig. 15. The control diagram is shown in Fig. 16. The parameters are the same with VSC II, where the grid-side inductor L_2 is 3 mH and the filter capacitor C is 5 μ F. Following the steps of the proposed method, the implementation of the proposed strategy for impedance identification of the active damping controlled VSC is divided into the offline step and online step. In the offline phase, first we initialize an ANN

TABLE II
STABILITY PREDICTION AT DIFFERENT OPERATING POINTS

Symbol	I_d	I_q	V_d	Stability
a	18 A	0 A	311 V	NO
b	15 A	8 A	311 V	YES

with the same structure shown in case study A. Then, we use the simulation model of VSC I to generate the admittance data, then after the training by feeding the simulation data into the structured ANN, the offline admittance model is gained. In the online phase, the dataset is gained through the field measurement, then the dataset is fed into the trained offline admittance model, which is trained with the transfer learning technique. After the training, the online impedance identification results are shown in Fig. 17. The errors between impedance identification and the measured data are shown in Fig. 18, which are less than 1%. Hence, the comparison shows that the proposed method is effective in the online impedance identification of the VSC even with the unknown control structure. Thus, we can use offline training by simulating a VSC with standard PI control structure, to identify a real VSC with the unknown control structure with online measurement of the real VSC.

C. Application of Generated Impedance Model and Experiment Validation

The generated MOP-impedance model with the proposed physics-informed neural network based online impedance identification is further used for stability analysis. The stability analysis is performed by the GNC [8]. The stability state of the converter-grid interaction system at the specific operating point is predicted by analyzing the eigenlocus. By sweeping the operating point of the MOP-impedance model, the stability state at the global operating points could be obtained.

The parameters of the target VSC in experiment is the same with the VSC II shown in Table I. Two cases of different point are tested. If the grid inductor is 18 mH, the Nyquist diagram of the two cases can be shown in Fig. 19. It can be easily seen that case a is unstable while case b is stable. The operating point and stability of case a and case b are shown in Table II.

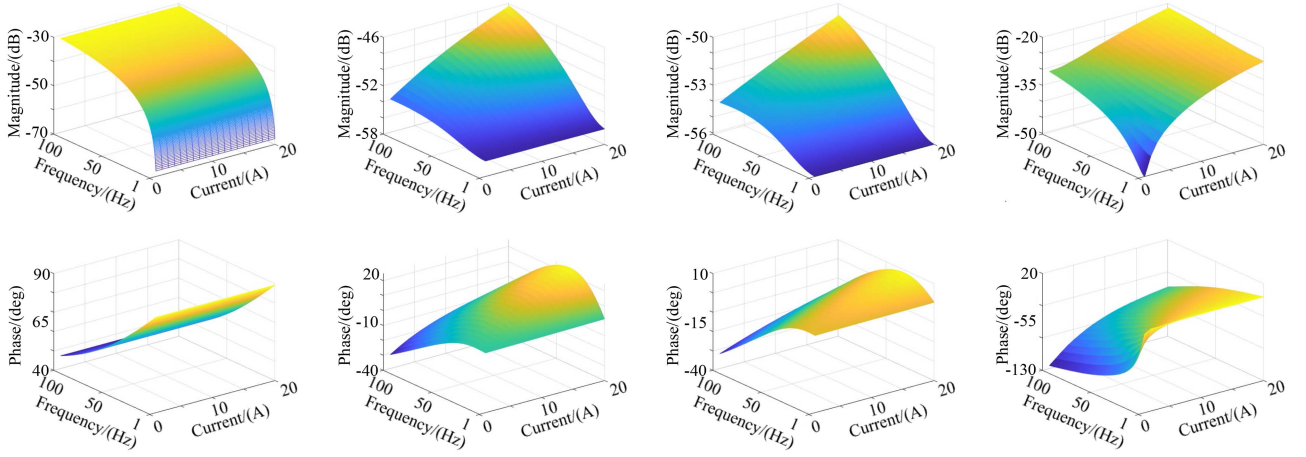


Fig. 17. Generated MOP-admittance model of active damping controlled VSC.

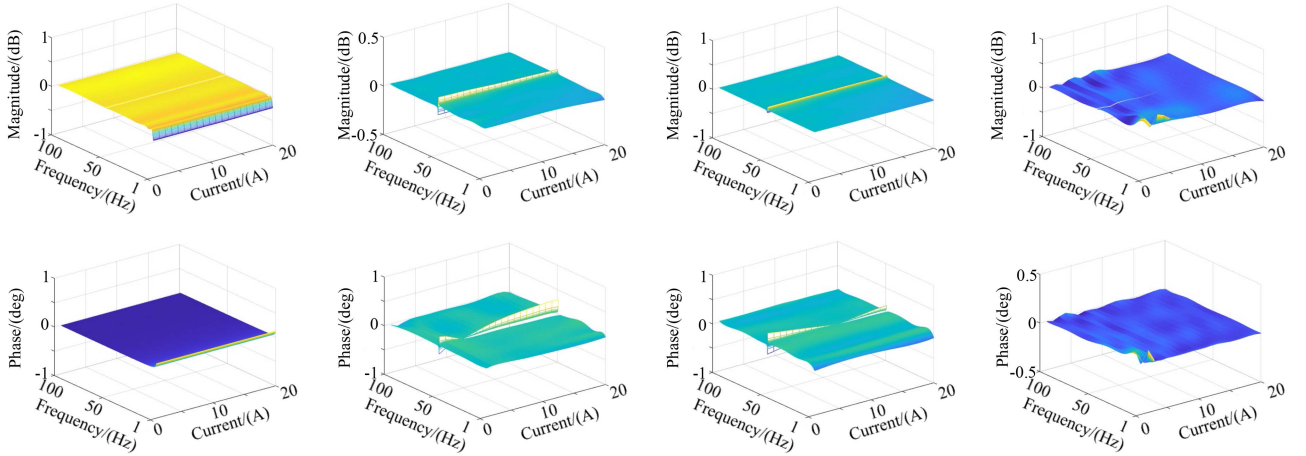


Fig. 18. Errors between generated admittance model using the proposed method and measured data of active damping controlled VSC.

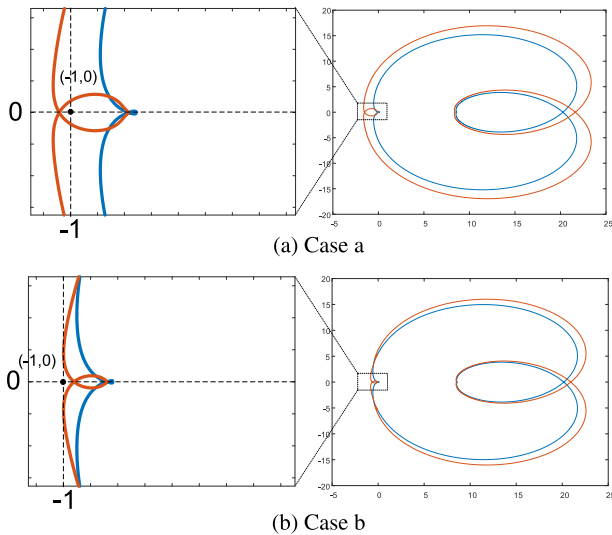


Fig. 19. Nyquist diagram of case a and case b.

To validate the stability analysis result, the experiment is conducted. The experimental tests setup is shown in Fig. 20. The dSPACE DS1007 is used to control the VSC. The Chroma

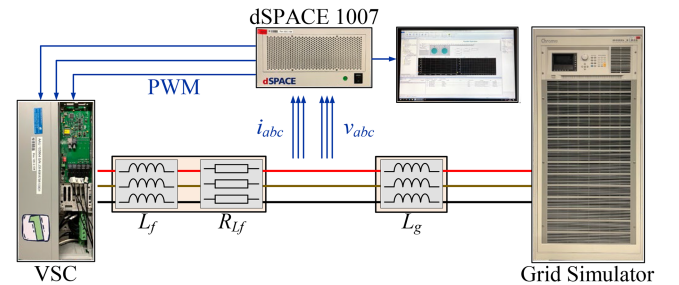


Fig. 20. Experimental setup.

61845 serves as the grid simulator. Two cases shown in Table II are tested, the corresponding experimental results are shown in Fig. 21, where V_{ga} represents the grid voltage and I_a , I_b , and I_c are the output current of VSC. When the VSC is at operating point a, the grid-connected VSC system is unstable, whereas the system is stable when the system is at the operating point b. The experimental results verify the accuracy of the admittance model generated with the proposed method, which shows the effectiveness of the proposed approach in impedance identification of VSC.

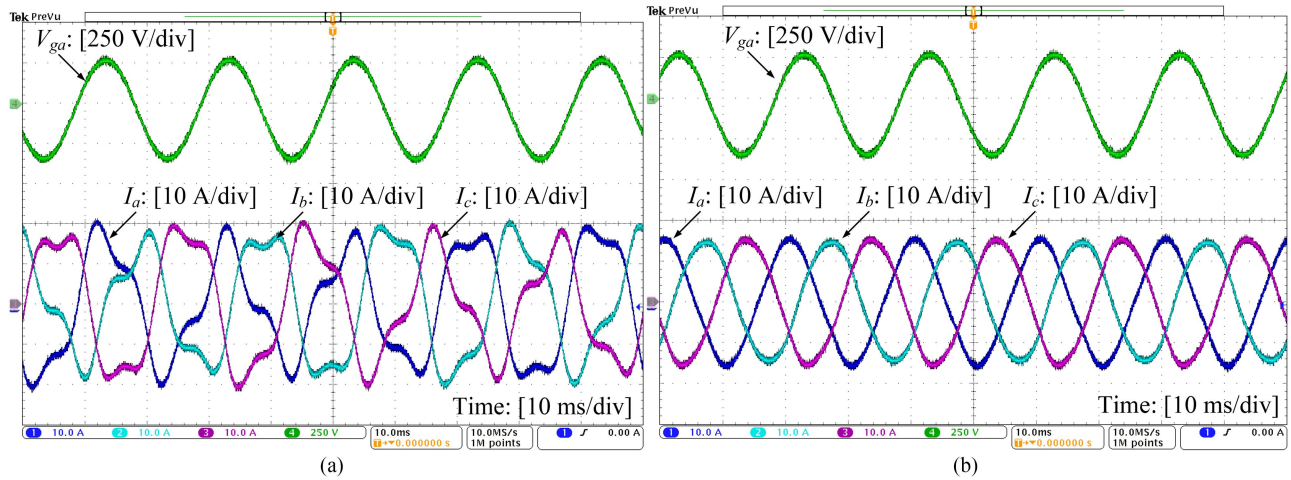


Fig. 21. Experimental results of grid-converter interaction system at different operating points. (a) Case I: operating point a. (b) Case II: operating point b.

V. CONCLUSION

This article proposed a physics-informed neural network based impedance identification method for online impedance identification of VSC under changing operating points. Considering the data amount limitation in online impedance identification, the physics-informed neural network based impedance identification was developed in this article. The partial physics of the VSC was used to compress the ANN, which can reduce the calculation burden of online impedance identification. Meanwhile, the two-steps impedance identification was developed with the inspiration of the transfer learning theory to further increase the online impedance identification efficiency. By making full use of partial physics and simulation data from a general VSC with the standard PI controller, the proposed method can enable an accurate online impedance identification of VSC with different and unknown control structures. The experimental results can validate that the proposed method can be used for online impedance identification and stability analysis effectively. The comparison results can verify the superiority of the proposed method to existing methods.

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Mengfan Zhang (Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2015 and 2018, respectively, and the Ph.D. degree in power electronic engineering from Aalborg University, Aalborg, Denmark, in 2022.

He was a Guest Ph.D. student with KTH, Stockholm, Sweden, from November 2020 to May 2021. He is currently a Postdoctoral Researcher with the KTH Royal Institute of Technology, Stockholm. His research interests include the modeling and stability analysis of the power electronic based power systems.



Qianwen Xu (Member, IEEE) received the B.Sc. degree from Tianjin University, Tianjin, China, in 2014, and the Ph.D. degree from Nanyang Technological University, Singapore, in 2018, both in electrical engineering.

She was a Postdoc Research Fellow with Aalborg University, Aalborg, Denmark, and a Wallenberg-NTU Presidential Postdoc Fellow with Nanyang Technological University during 2018–2020. She was also a Visiting Researcher with Imperial College London, London, U.K., from March 2020 to June 2020. She is currently an Assistant Professor with the Department of Electric Power and Energy Systems, KTH Royal Institute of Technology, Stockholm, Sweden. Her research interests include advanced control, optimization, and AI application for microgrid and smart grid.

Dr. Xu serves as the Vice Chair for the IEEE Power and Energy Society and Power Electronics Society, Sweden Chapter, and an Associate Editor for the IEEE TRANSACTIONS ON SMART GRID and IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS. She was a recipient of Humboldt Research Fellowship, Excellent Doctorate Research Work in Nanyang Technological University, Best Paper Award in IEEE PEDG 2020, etc.



Xiongfei Wang (Senior Member, IEEE) received the B.S. degree in electrical engineering from Yanshan University, Qinhuangdao, China, in 2006, the M.S. degree in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2008, and the Ph.D. degree in energy technology from Aalborg University, Aalborg, Denmark, in 2013.

Since 2009, he has been with the Department of Energy, Aalborg University (AAU Energy), where he became an Assistant Professor in 2014, an Associate Professor in 2016, and a Professor and Leader of Electronic Power Grid (eGRID) Research Group in 2018. He has also been a part-time Professor with the KTH Royal Institute of Technology, Stockholm, Sweden, since 2020. His current research interests include modeling and control of power electronic converters and systems, stability and power quality of power-electronics-dominated power systems, and high-power converters.

Dr. Wang serves as the Co-Editor-in-Chief for the IEEE TRANSACTIONS ON POWER ELECTRONICS and an Associate Editor for the IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS (JESTPE). He was a recipient of nine Prize Paper Awards in the IEEE Transactions and Conferences, the 2016 AAU Talent for Future Research Leaders, the 2018 Richard M. Bass Outstanding Young Power Electronics Engineer Award, the 2019 IEEE PELS Sustainable Energy Systems Technical Achievement Award, the 2020 IEEE PES Prize Paper Award, the 2020 JESTPE Star Associate Editor Award, the 2022 Isao Takahashi Power Electronics Award, and the Highly Cited Researcher in the Web of Science from 2019.